

EE 3150: Signals and Systems

Lab 2: Impulse Response, Step Response, and RC Circuits

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In this lab, you will work in your groups (same partner as before) to:

- Understand how a short pulse can be used to approximate the *impulse response* $h(t)$ of a linear system, and how short the pulse must be for the approximation to be accurate.
- Derive the pulse and step responses of a two-stage RC cascade analytically (by hand).
- Numerically obtain the impulse and step responses of the RC circuit in Python.
- Build a two-stage RC circuit and measure its impulse, step, and frequency responses on the oscilloscope, then compare measurement to theory¹.

This lab has a prelab component that you must complete before your lab session, and an in-lab component done during the session. The prelab is **Part 1** and **Part 2**; the in-lab work is **Part 3**. You must show your completed **Part 1** to the TA before you are allowed to start Part 3 (this is the “gate”), so make sure Part 1 is done fully *before* your session. Part 2 is also required, but the TA will not check it at the start of lab. Work through the entire lab (all three parts) as a team and collect data together; you may split the work on report writing. You will submit one report per team.

1 Part 1: Background and Analysis

1.1 Impulse response from a narrow pulse

Consider the series RC circuit of Fig. 1, with output v_c being the voltage across the capacitor. The capacitor has capacitance C and the resistor has resistance R . The input voltage is the signal $v_0(t)$. The governing ODE of this circuit (with $\tau = RC$ denoting the time constant) is

$$\tau \dot{v}_c(t) + v_c(t) = v_0(t), \quad v_c(0) = 0. \quad (1)$$

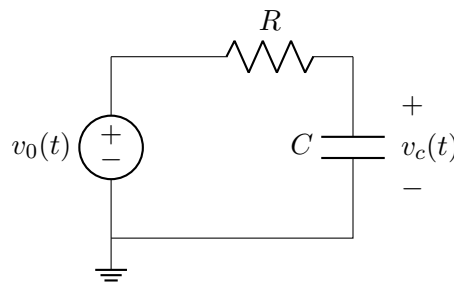


Figure 1: Series RC circuit.

¹This lab is based on materials by Profs. Kay and Duffaut; polished with the help of Anthropic Claude.

We would like to find the impulse response, $h(t)$, of this system. Recall the impulse response is the system output when the input is a Dirac delta function $\delta(t)$. However, $\delta(t)$ is a mathematical construct and is not physically realizable. Instead, we approximate it by using a narrow rectangular pulse of width T and unit area: $v_0(t) = \delta_T(t)$, where

$$\delta_T(t) = \begin{cases} 1/T, & 0 < t < T, \\ 0, & \text{otherwise.} \end{cases}$$

To find v_c , we solve the ODE by hand. To this end, we consider the response during and after the pulse. During the pulse, multiply (1) by the integrating factor $\mu = e^{t/\tau}$ and solve:

$$v_c(t) = \frac{1}{T}(1 - e^{-t/\tau}), \quad 0 \leq t < T,$$

where $v_c(0) = 0$ fixes the constant of integration. After the pulse, the input is zero, so the capacitor discharges from $v_c(T)$ as $v_c(t) = v_c(T) e^{-(t-T)/\tau}$. Collecting both intervals,

$$v_c(t) = \begin{cases} \frac{1}{T}(1 - e^{-t/\tau}), & 0 \leq t < T, \\ \frac{1}{T}(e^{T/\tau} - 1) e^{-t/\tau}, & t \geq T. \end{cases} \quad (2)$$

As the pulse becomes infinitely narrow ($T \rightarrow 0^+$), the output, by definition, converges to the impulse response. Since we are interested in $t > 0$, we focus on the second line of (2). We note that $\lim_{T \rightarrow 0^+} (e^{T/\tau} - 1)/T = 1/\tau$ by l'Hôpital's rule, so (2) converges to the **impulse response**:

$$h(t) = \frac{1}{\tau} e^{-t/\tau} u(t) \quad (3)$$

where $u(t)$ is the unit step function, and including it simply tells us that $h(t) = 0$ for $t < 0$. A key fact you will verify in simulation: as $T \rightarrow 0$ the response depends only on the pulse *area*, not its shape: a triangular unit-area pulse yields the same $h(t)$ in the limit.

1.2 How short is “short”?

We cannot take $T \rightarrow 0$ in practice: an infinitely narrow, unit-area pulse would need infinite height. So we use a small but finite T and ask how much error that introduces. Compare the ideal impulse response $h(t) = \frac{1}{\tau} e^{-t/\tau}$ in (3) with the finite-pulse response (2) for $t \geq T$: the two share the same shape $e^{-t/\tau}$ but differ in amplitude. So, introduce their ratio, $K(T)$ and call it the “correction factor”:

$$K(T) = \frac{\text{finite-pulse amplitude}}{\text{ideal amplitude}} = \frac{(1/T)(e^{T/\tau} - 1)}{1/\tau} = \frac{e^{T/\tau} - 1}{T/\tau}.$$

It measures how faithfully the finite pulse reproduces the impulse response: $K(T) = 1$ would be exact. To see how K departs from 1 when T is small, let $x = T/\tau$ and apply Taylor-series expansion of the exponential, $e^x = 1 + x + \frac{x^2}{2} + \dots$, so that

$$K(T) = \frac{e^x - 1}{x} = 1 + \frac{x}{2} + \frac{x^2}{6} + \dots = 1 + \frac{T}{2\tau} + \dots$$

The finite pulse therefore overestimates the ideal amplitude by a relative error $\approx T/(2\tau)$. Requiring this error to be at most 1% gives the design rule

$$T \lesssim 0.02 \tau \quad (\text{for } 1\% \text{ error}).$$

For a higher order system with multiple time constants (e.g., the two-stage RC circuit below), the rule applies to the *smallest* time constant (i.e., the fastest dynamics), though the error bound above would now be approximate instead of exact.

1.3 The lab circuit: two-stage RC network with gain

The system you will build and simulate (Fig. 2) is an RC circuit, an OP-AMP buffer with a fixed gain G , and a second RC circuit in cascade.

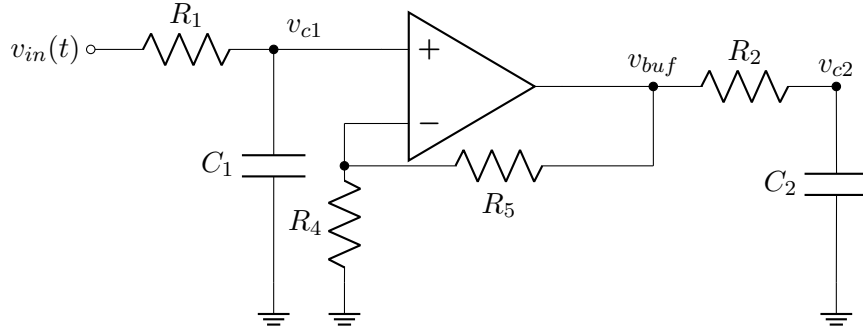


Figure 2: Two-stage RC cascade: an input RC circuit, a non-inverting amplifier, and an output RC circuit.

Each RC circuit, in isolation, behaves like the one in the previous section. If we put them in series without an OP-AMP stage in between, the second RC stage pulls current from the first stage. The goal of the buffer is to isolate the stages and prevent this loading effect. This makes both analysis and design much easier. Writing KCL at the negative terminal of the OP-AMP, we get: $v_{buf} = G v_{c1}$, where

$$G = 1 + \frac{R_5}{R_4}$$

is the gain of the OP-AMP stage. The first RC stage sees v_{in} as its input and produces v_{c1} , which is then amplified by the OP-AMP to produce v_{buf} . The second RC stage takes v_{buf} as its input and produces the final output v_{c2} (see, the OP-AMP made the analysis easier). Writing the ODE for each RC stage, we thus obtain:

$$\begin{aligned} \dot{v}_{c1} &= \frac{1}{\tau_1} (v_{in} - v_{c1}), \quad \tau_1 = R_1 C_1 \\ \dot{v}_{c2} &= \frac{1}{\tau_2} (G v_{c1} - v_{c2}), \quad \tau_2 = R_2 C_2. \end{aligned} \tag{4}$$

For this lab, we will use the following parameters:

$$R_1 = 10 \text{ k}\Omega, C_1 = 1 \text{ }\mu\text{F};$$

$$R_2 = 4.7 \text{ k}\Omega, C_2 = 1 \text{ }\mu\text{F};$$

$$R_4 = 1 \text{ k}\Omega, R_5 = 4.7 \text{ k}\Omega.$$

Task 1

Answer the following and include your answers (and full derivations) in your report:

1. Using the parameters above, compute the time constants τ_1 and τ_2 and the gain G . Write your results in the report.
2. Derive the equivalent second-order ODE for $v_{c2}(t)$ in terms of $v_{in}(t)$, τ_1 , τ_2 , and G . Do this by differentiating both sides of the second equation and substituting $\dot{v}_{c1}$ from the first equation.
3. Derive the impulse response $h(t)$ of the cascade. Do this by solving the first order ODE (the solution is similar to the previous subsection) and plugging the solution into the second ODE. Then, solve the second ODE to get $h(t)$. Finally, plot $h(t)$ vs. t .
4. Recall the step response, $s(t)$, is defined as the response to the unit step input. Equivalently, we can express $s(t)$ in terms of $h(t)$: $s(t) = \int_0^t h(s) ds$. Using the impulse response from the previous part, compute $s(t)$.
5. The final value of $s(t)$ (i.e., $\lim_{t \rightarrow \infty} s(t)$) is referred to as the DC gain. Find it as a function of circuit parameters.
6. Show that the v_c response to a rectangular pulse of height A and width T follows by superposition:

$$v_{out}(t) = A[s(t) - s(t - T)]. \quad (5)$$

7. Use the above to find $v_{out}(t)$ using $s(t)$ from the previous part.
8. Using the results of Section 1.2, determine how large T can be for v_{out} to approximate $h(t)$ to within 1% error.

In the previous section, we introduced a pulse of height $1/T$ and width T to approximate the impulse. This is fine when we are just dealing with mathematical interpretations, but in engineering we measure things and need to be mindful of the practicalities of the measurement. Let's say we decide we need a pulse width of $1 \mu s$ then we would need a voltage level of 1 Megavolt to get a unit pulse. This is clearly not practical for our circuit, so the pulses we use will not have amplitude $1/T$, but something smaller (A in Task 1 above). This results in a pulse area of $A \times T$, which will be smaller than 1. Good news is that this is not an issue: to convert the experimentally measured response $v_{out}(t)$ (resulting from a pulse of height A and width T) into the theoretical unit impulse response $h(t)$, we divide the measured signal by the total pulse area $A \cdot T$ and write: $h(t) \approx \frac{v_{out}(t)}{A \cdot T}$.

Deliverable

Answers to the questions in Task 1 above (each part is worth 1 point except part 3 which is worth 2 points). Show all your work and derivations in your report. **You must show your answers to the TA at the start of the lab before proceeding to the hardware experiment. Having done Part 1 before the lab session is worth 5 points.**

2 Part 2 — Numerical calculation of impulse and step responses in Python

You will simulate the cascade (4) numerically and compare it against the closed-form $h(t)$, $s(t)$, and pulse response derived above. You will carry out three experiments that build on one another: rectangular pulses of shrinking width (the response theoretically approaches $h(t)$); repeat with a triangular pulse to show that only the pulse *area* matters, not its shape; and a step (the step response and its DC gain). You will be given a starter code framework.

Important note: You will use `scipy.integrate.solve_ivp()` to solve the ODE numerically. This solver is a Runge-Kutta solver with adaptive step size. It is robust and accurate, but you must be careful to set the maximum step size appropriately for the problem at hand. In your case, the circuit is driven by pulses as short as $100\mu\text{s}$ while its slowest time constant is 10ms . An adaptive ODE solver left to itself can step *over* a narrow pulse and integrate almost no input. Always cap the solver's `max_step` to a small fraction of both the pulse width and the fastest time constant. This has been done for you below.

Task 2.1: Python framework

Create lab2.py with the code below. It defines the component values, the cascade ODE, the analytic $h(t)$ and $s(t)$, and an ODE solver.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.integrate import solve_ivp
4
5 R1, C1 = 10e3, 1e-6; tau1 = R1*C1
6 R2, C2 = 4.7e3, 1e-6; tau2 = R2*C2
7 R4, R5 = 1e3, 4.7e3; G = 1 + R5/R4
8 tau_min, tau_max = min(tau1, tau2), max(tau1, tau2)
9
10 A = 1.0
11 T = 50e-3
12 t_end = T + 8*tau_max
13 t_eval = np.linspace(0.0, t_end, 4000)
14 max_step = min(T, tau_min)/20.0
15
16 vin = np.zeros(len(t_eval))
17 vin[(t_eval >= 0) & (t_eval <= T)] = A
18
19 def h(t):          # returns impulse response, t>=0
20     t = np.asarray(t, float);
21     y = np.zeros_like(t);
22     y = G/(tau1-tau2)*(np.exp(-t/tau1) - np.exp(-t/tau2))
23     y[t<=0] = 0.0
24     return y
25
26 def s(t):          # returns step response, t>=0
27     t = np.asarray(t, float);
28     y = np.zeros_like(t);
29     y = G/(tau1-tau2)*(tau1*(1-np.exp(-t/tau1)) - tau2*(1-np.exp(-t/tau2)))
30     y[t<=0] = 0.0
31     return y
32
33 def f_rhs(t, x):    # needed for solve_ivp
34     v1, v2 = x
35     if (0.0 <= t < T):
36         _vin = A
37     else:
38         _vin = 0.0
39     return [(_vin - v1)/tau1, (G*v1 - v2)/tau2]
40
41 sol = solve_ivp(f_rhs, (0.0, t_end), [0.0, 0.0], t_eval=t_eval, rtol=1e-8, atol=1e-8,
42                 max_step=max_step)
43
44 v2 = sol.y[1] # pulse response
45 v_analytic = A*(s(t_eval) - s(t_eval - T)) #theoretical pulse response adjusted by A
```

Task 2.1: continued

Now use the code below to plot the results. The first figure shows the input and output for the rectangular pulse. The second figure overlays the area-normalized output with the analytic impulse response $h(t)$.

```
1 fig1, ax = plt.subplots(2, 1, figsize=(8, 6))
2 fig2, axn = plt.subplots(figsize=(8, 4.2))
3 axn.plot(t_eval * 1e3, h(t_eval), "k", lw=2, label="h(t) analytic")
4
5 ax[0].plot(t_eval * 1e3, vin)
6 ax[1].plot(t_eval * 1e3, v2)
7 axn.plot(t_eval * 1e3, v2 / (A * T), "--")
8
9 ax[0].set(title="Rectangular inputs v_in(t)", ylabel="V"); ax[0].set_xlim(0, 80)
10 ax[1].set(title="Outputs v_c2(t)", xlabel="t (ms)", ylabel="V"); ax[1].set_xlim(0,
11         80)
12
13 axn.set(title="h(t) and its approximation",
14         xlabel="t (ms)", ylabel="V"); axn.set_xlim(0, 60)
15 axn.grid(True, alpha=.4); axn.legend(fontsize=8)
16 plt.show()
```

Run the code and make sure you get the plots. You will modify this code in the next tasks to simulate different pulse widths and shapes.

Task 2.2 — Rectangular pulses

Simulate the response to rectangular pulses of amplitude $A = 1\text{ V}$ and widths $T \in \{100\text{ }\mu\text{s}, 1\text{ ms}, 10\text{ ms}, 50\text{ ms}\}$. Run the code with each T and produce the plots. Include the plots in your report and interpret your observations. In particular, comment on which T values collapse onto $h(t)$ and which do not, and relate this to the rule $T \lesssim 0.02 \tau_{\min}$.

Then repeat with $A = 5\text{ V}$, $T = 100\text{ ms}$ and compare its $v_{c2}/(AT)$ plot with the $A = 1\text{ V}$, $T = 100\text{ ms}$ case. Include the plot in your report. Do the two A values at $T = 100\text{ ms}$ give identical area-normalized curves? What property of the system does this demonstrate? Include your answer in the report.

Task 2.3 — Triangular pulse

Now modify the Python code (where `vin` and `_vin` are defined) to simulate the response to a triangular pulse of width $T = 100\text{ }\mu\text{s}$ and peak $A = 2\text{ V}$, whose area $\frac{1}{2}AT = 1 \times 10^{-4}\text{ V s}$ equals that of the $1\text{ V}/100\text{ }\mu\text{s}$ rectangle.

Plot the triangular input and overlay $v_{c2}/(0.5AT)$ against $h(t)$ (note: you need to modify the plotting code to normalize correctly). Include your plot in your report, and state whether the pulse *shape* matters when the area is the same. Mathematically, explain why. Include the parts of the code you modified in your report.

Task 2.4 — Step response

Now modify the code to simulate the step response for $A = 1$ V and overlay the analytic $s(t)$. Do this by letting v_{in} be a step of height A instead of a pulse. Include the plot of $s(t)$ in your report.

Once you calculate the step response, differentiate it to get the impulse response and compare it with the analytic $h(t)$. Include your plot that overlays $h(t)$ and $\frac{ds(t)}{dt}$ in your report and explain whether $h(t)$ and $ds(t)/dt$ are the same. Note: to differentiate the step response numerically, you can use the `np.gradient()` function in Python. Include the code snippet you modified in your report.

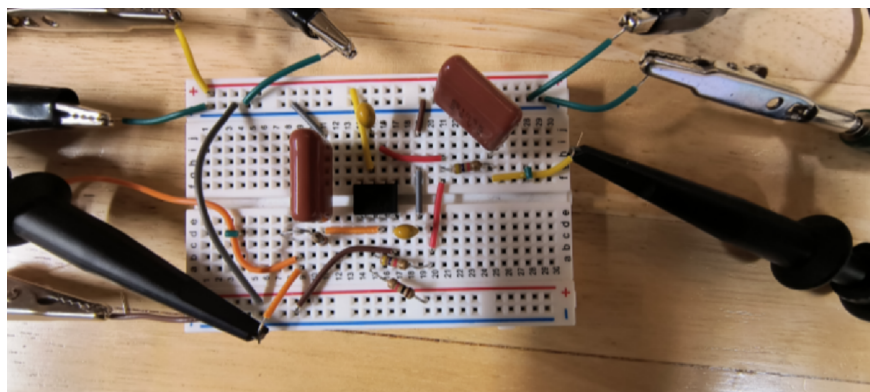
In your report, confirm the final value equals G , and comment on the settling time in terms of τ_1 and τ_2 .

Deliverable

Include in your report the requested code, plots, explanations, and derivations in Tasks 2.2 [6 pts], 2.3 [4 pts], and 2.4 [5 pts]. Include the code snippets that you modified (do not include the entire code).

3 Part 3 — Lab Procedure

You will now build the circuit of Fig. 2 and measure its impulse, step, and frequency responses. This assumes familiarity with the equipment in V334. The OP-AMP you will use is a LM-741 OP-AMP (data sheet attached). The circuit is powered by a dual supply at ± 15 V, and the function generator is used to provide the input signal. The oscilloscope is used to measure both the input and output signals. Make sure to use a 0.01 μ F ceramic capacitor to bypass the op-amp power supply pins (standard practice; capacitors not shown in Fig. 2). The other capacitors, C_1 and C_2 , are polyester capacitors. The resistors are standard 1/4 W resistors. See the picture below for a reference of the circuit layout on a protoboard. The layout is not unique, but it is important to keep the leads short and the bypass capacitors close to the op-amp.



Wear safety glasses at all times. And no food/drinks in the lab. No exceptions.

Task 3.1 — Build and instrument

Build the circuit on a protoboard, noting that the layout is not unique. Keep wires and leads short.

- You will use a dual supply at $\pm 15\text{ V}$ (common ground), current-limited to $\leq 0.1\text{ A}$, to power the OP-AMP. Ask the TA if you do not know how to do this.
- You will use a function generator to generate v_{in} . Set the generator output impedance to *High Z*.
- Your scope Ch1 will be v_{in} and Ch2 $\rightarrow v_{c2}$.
- Scope Ch1 probe $\times 10$ (scope setting also $\times 10$); Ch2 probe $\times 1$ (scope also $\times 1$, since v_{c2} is small and low-frequency). Set both channels to **DC coupling**. Trigger on Ch1. Set the vertical scale of channel 1 to 2 V/div . Set the vertical scale of channel 2 to 100 mV/div . Note: The scope probe frequency response is much lower on $\times 1$ so it is standard practice to use probes on $\times 10$ unless the signal is quite small. In this case the signal for channel 2 is small and the frequency content relatively low so $\times 1$ is the correct choice. This is an exception and you should use $\times 10$ in your normal lab work.

Task 3.2 — Impulse-response (pulse) measurements

Turn on the $+15\text{ V}$ and -15 V power supply. Set the function generator to output a pulse train with the following parameters: amplitude 5 V , offset 2.5 V (so the pulse swings $0 \rightarrow 5\text{ V}$), width $100\text{ }\mu\text{s}$, frequency 10 Hz . Turn on the function generator output and adjust the scope trigger until you get a stable trace. Then save **both** a **.png** screenshot and a **.csv** data file — once on a slow sweep ($\sim 20\text{ ms/div}$) showing the full decay, and once on a fast sweep ($\sim 100\text{ }\mu\text{s/div}$) showing the response during and just after the pulse. You should be able to see two very narrow input pulses spaced 100 ms apart on channel 1 and a much wider impulse response on channel 2. Show your traces to the TA to receive in-lab credit.

Repeat (adjusting vertical/horizontal scales for a good trace, saving a fast and a slow capture each) for:

- amplitude 5 V , offset 2.5 V , width 1 ms , 10 Hz ;
- amplitude 2 V , offset 1 V , width 10 ms , 10 Hz ;
- amplitude 2 V , offset 1 V , width 50 ms , 1 Hz (the slow sweep takes a while to paint).

Take your **.png** and **.csv** files. Take the time to organize so you can find them later and do not get confused.

In your report: include all four plots (one from the first paragraph above, and three from the three bullets). Then, normalize each of the four v_{c2} traces by the pulse area like you did in Part 2, and plot the normalized $v_{c2}(t)$ against the analytic $h(t)$ on one figure. Interpret your results and comment on the agreement between the four normalized traces and $h(t)$ and reasons for any disagreement.

Task 3.3 — Step-response measurements

A step gives a larger, easier-to-measure output carrying information that can be used to compute $h(t)$ (by differentiation). Set the function generator to output a pulse with the following parameters: Amplitude: 500mV, Offset: 250mV, Pulse width: 100ms (i.e., large width), Frequency: 5 Hz, and Output: Hi Z. Set the scope to trigger on channel 1. Set the vertical scale of channel 1 to 200mV/div. Set the vertical scale of channel 2 to 1V/div. Set the horizontal scale to 50ms/div. Turn on the function generator and adjust the scope for a stable trace. You should see around 2 or 3 pulses on the input along with the output response. Save .png and .csv files of the traces. You should see the rising edge and falling edge responses have the same time constants.

Now increase the amplitude to 5V and the offset to 2.5V. Save .png and .csv files of the traces. You will notice that the rising edge and falling edge responses are not the same in this case.

In your report: Include the two plots. Also, note the asymmetry between rising and falling edges of the large step and give a physical cause (e.g. op-amp slew/saturation, supply limits, probe loading).

Task 3.4 — Frequency response measurements

You will now drive the same circuit with sinusoidal inputs of various frequencies and measure how each stage attenuates and delays them. The data you gather here will be later compared with analytical derivations in Labs 3 and 4 so keep your data.

Set the function generator to a *sine* output (instead of pulse) at $V_{in} \approx 1$ V peak-to-peak and 0 DC value. For each frequency $f \in \{5, 15, 30, 60, 120\}$ Hz: put v_{in} on Ch1 and v_{c2} on Ch2, adjust the scales for a clean trace (two cycles on screen), and read the peak-to-peak amplitude and the time shift Δt between output and input. Repeat with Ch2 on v_{c1} . Fill one row of the table below per frequency. Note: use the oscilloscope cursors to help you with the measurements.

f (Hz)	V_{c1} (pp)	Δt_1 (s)	V_{c2} (pp)	Δt_2 (s)
5				
15				
30				
60				
120				

Double check your measurements and take the table home with you. As a check before you leave the bench, confirm that at $f = 5$ Hz you get $V_{c2} \approx G$ and $V_{c1} \approx 1$ peak-to-peak; if either is far off, re-check your work.

In your report: include the completed table; make one magnitude plot showing both v_{c1} and v_{c2} versus frequency; state which stage rolls off faster with frequency.

Deliverable

Showing your working circuit to the TA is worth [5pts]. In your reports, plots and answers to the questions in Task 3.2 are worth [8 pts]. Plots and answers to the questions in Task 3.3 are worth [4 pts]. Task 3.4 is worth [6 pts], 3 points for the table, 2 points for the plot, and 1 point for the comparison.

Submission

Submit **one** typed PDF per group via Brightspace before the deadline, plus your Python code. Make sure all plots are properly labeled. Style of your report, including figures carries [4pts]. Include an **introduction** [1pt] stating the objectives and a **conclusion** [3pt] summarizing the results and, more importantly, explaining what you learned and take-away messages. Be thorough with your conclusion to receive full credit. The report must be typed, except the hand-drawn figures, which could be included in the report as pictures. Completing the pre-lab before the lab session and showing your scope to your TA during the lab session is worth [5 pts]. **The lab is graded out of 60 points**



LM741 Operational Amplifier

1 Features

- Overload Protection on the Input and Output
- No Latch-Up When the Common-Mode Range is Exceeded

2 Applications

- Comparators
- Multivibrators
- DC Amplifiers
- Summing Amplifiers
- Integrator or Differentiators
- Active Filters

3 Description

The LM741 series are general-purpose operational amplifiers which feature improved performance over industry standards like the LM709. They are direct, plug-in replacements for the 709C, LM201, MC1439, and 748 in most applications.

The amplifiers offer many features which make their application nearly foolproof: overload protection on the input and output, no latch-up when the common-mode range is exceeded, as well as freedom from oscillations.

The LM741C is identical to the LM741 and LM741A except that the LM741C has their performance ensured over a 0°C to +70°C temperature range, instead of –55°C to +125°C.

Device Information⁽¹⁾

PART NUMBER	PACKAGE	BODY SIZE (NOM)
LM741	TO-99 (8)	9.08 mm × 9.08 mm
	CDIP (8)	10.16 mm × 6.502 mm
	PDIP (8)	9.81 mm × 6.35 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

Typical Application

